

Assignment no. 1

Q-1) Task assignment approach with example:
Ans Task assignment is the process of distributing tasks or jobs among multiple nodes in a distributed system so that execution becomes faster, balanced and efficient.

Working:

In a large distributed system:

- A large task is divided into smaller subtasks.
- A scheduler decides which node will execute which task.
- Tasks are assigned based on load, capability, and availability.
- Node executes tasks in parallel.
- Results are collected and combined to produce final output.

Task assignment can be:

- Static - Decided before execution.
- Dynamic - Decided during execution or based on system state.

Example:

Consider a video streaming platform (like youtube):

- When many users upload videos, the system divides work into tasks:

- i.) Video compression
 - ii.) Thumbnail generation
 - iii.) Format conversion
- Different servers handles different tasks simultaneously.

Number of tasks

Routing System verifies time necessary for the fulfillment of task

task assignment module

Estimated number optimal AGV's

Assignment of tasks for AGV's

Routing System simulates routes for the AGV's

Does the estimated model reflects the simulated model

No

yes

Tasks and routes are sent to the AGV's

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Figure shows Task assignment module architecture

Advantages:

- 1.) Improve overall system performance.
- 2.) Enables parallel processing.
- 3.) Reduces execution time.
- 4.) Balances load across nodes.
- 5.) Efficient resource utilization.
- 6.) Supports high scalability.

Disadvantages:

- 1.) Complex scheduling algorithm required.
- 2.) Communication overhead between nodes.
- 3.) Task dependency may cause delay.
- 4.) Load imbalance can still occur.
- 5.) Difficult fault handling.
- 6.) Requires synchronization mechanisms.

Q-2) Explain code mitigation:

Ans Definition:

Code mitigation is the movement of code from one machine to another in a distributed system to execute it closer to resource.

Diagram:

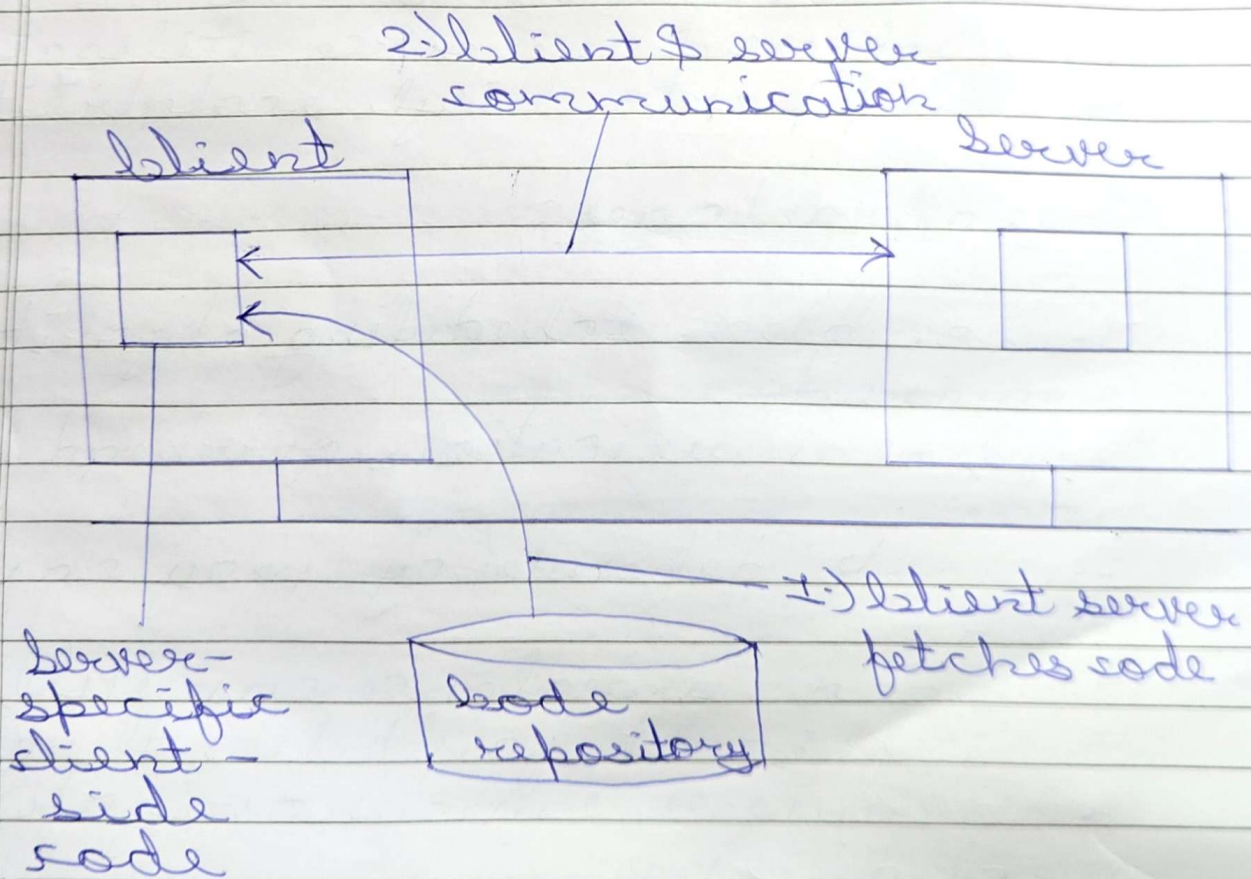


Figure shows code mitigation:

Working:

- Code is packed and transferred over the network.
- It is executed on mobile system.
- It may carry its execution state (strong migration) or just code (weak migration).
- Results are sent back or further processed.

Example:

- In an e-commerce platform:
- Instead of sending huge customer data to one server.
 - A mobile agent (code) move to different servers.
 - It processes data locally (e.g. recommendation generation).
 - Returns only final results → reduces network traffic.

Advantages:

- 1-) Reduces network load.
- 2-) Faster processing near data.
- 3-) It improves efficiency.

- 4.) Supports dynamic systems.
- 5.) Better resource utilization.
- 6.) Enables mobile computing.

Disadvantages:

- 1.) Security risk (malicious code execution).
- 2.) Platform compatibility issue.
- 3.) Complex implementation.
- 4.) Difficult in debugging.
- 5.) Execution overhead is very high.
- 6.) Requires proper runtime environment.

Q.3.) Explain strict consistency and sequential consistency model?

Ans. A.) strict consistency model:

Definition:

Strict consistency ensures that every read operation returns the most recent write value instantly across all nodes.

Working:

- When a value is updated, all nodes immediately see the changes.
- Requires global synchronization and instant updates.
- No delay or stale data allowed.

Example:

In a real-time stock trading system:

- If a stock price changes, all users must see the latest price instantly.
- Any delay can cause financial loss.

Advantages:

- 1.) Provides highest consistency.
- 2.) No outdated data.
- 3.) It is very easy to understand.
- 4.) It gives accurate results.
- 5.) Reliable system behaviour.
- 6.) Suitable for critical execution.

Disadvantages:

- 1.) It has very high latency.

- 2.) It is not scalable.
- 3.) Difficult to implement.
- 4.) Requires strict synchronization.
- 5.) Cost required is high.
- 6.) Poor performance in distributed environments.

B.) Sequential consistency model:

Definition:

Sequential consistency ensures that all operations are executed in the same order for all processes, but not necessarily in real time.

Working:

- Operations are arranged in sequence.
- All nodes follow the same sequence.
- Timing may differ but order remains same.

Example:

In a collaborative document

editing system:

- If user A writes first and user B writes later.
- All users see edits in same order.
- Even if there is slight delay.

Advantages:

- 1.) Easier to implement than strict consistency.
- 2.) Maintains operation order.
- 3.) Predictable system behaviour.
- 4.) Better performance than strict consistency.
- 5.) It simplifies programming.
- 6.) It also ensures fairness.

Disadvantages:

- 1.) Not real-time accurate.
- 2.) Delays are possible.
- 3.) Ordering overhead is high.
- 4.) Synchronization is required in this model.
- 5.) Reduces performance due to overhead.
- 6.) Not suitable for time-critical systems.

Q-4-) Explain FIFO and weak consistency models.

Ans FIFO consistency model:

Definition:

FIFO consistency models guarantees that writes from a single process as seen by all other processes in the order they were used.

Working:

- Each process maintains its own order of operations.
- Messages from a sender are delivered in sequence.
- No guarantees for global order among different senders.

Example:

In a chat application (like WhatsApp):

- If a user sends message: "Hi"
- "How are you?"
- All receivers see them in that order.

same order.

- But messages from different users may arrive in different order.

Advantages:

- 1.) Maintains order per sender.
- 2.) Simple and efficient.
- 3.) It has very low overhead.
- 4.) Useful for messaging systems.
- 5.) Improves user experience.
- 6.) Easy to implement.

Disadvantages:

- 1.) No global ordering.
- 2.) Inconsistency across users.
- 3.) Not suitable for critical systems.
- 4.) It gives limited guarantees.
- 5.) Complex multi-user scenarios.
- 6.) There are many synchronization challenges.

Weak consistency model:

Definition:

Weak consistency model ensures that updates are visible only after synchronization operations, not immediately.

Working:

- Data updates are not instantly visible.
- Synchronization points ensure consistency.
- Before sync, different nodes may have different values.

Example:

In a distributed caching system:

- Data is updated in one server.
- Other server sees updated data only after cache refresh or sync.
- Until then, they may use old data.

Advantages :

- 1.) It gives high performance.
- 2.) Reduces communication overhead.
- 3.) It has scalable system.
- 4.) Efficient resource usage.
- 5.) Flexible implementation.
- 6.) Less synchronization cost.

Disadvantages :

- 1.) Stale or outdated data.
- 2.) Programmer must handle consistency.
- 3.) Difficult in debugging.
- 4.) Risk of incorrect results.

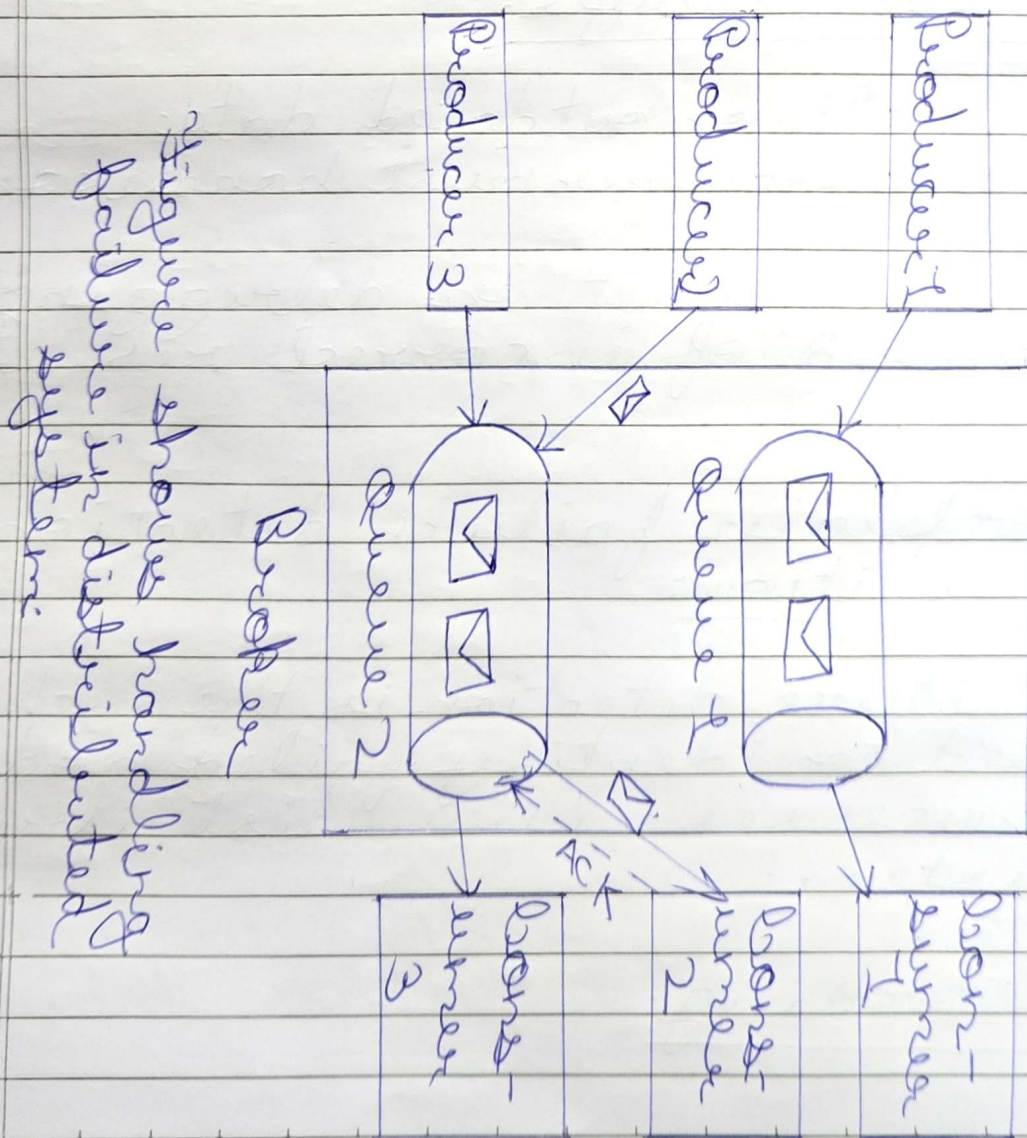
Q5.) Explain failure detection.
Ans Definition:

Failure detection is the mechanism used to identify failed nodes or processes in a distributed system.

Working:

- Nodes send periodic heartbeat signals.
- If a node does not respond within a time limit $\rightarrow$ considered failed.
- System may trigger recovery or reassign tasks.

Diagram:



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Example:

In a cloud computing platform (like AWS):

- If a server crashes.
- Monitoring system detects failure.
- Another server automatically takes over tasks.
- Ensure continuous service.

Advantages:

- 1.) It improves system reliability.
- 2.) Enables fault tolerance.
- 3.) Supports automatic recovery.
- 4.) It also maintains availability.
- 5.) Detects failure quickly.
- 6.) It helps in enhancing performance.

Disadvantages:

- 1.) False failure detection due to delay.

- 2.) Network latency issues.
- 3.) Extra communication overhead.
- 4.) It uses complex algorithms.
- 5.) High resource consumption.
- 6.) Detection delay possible.

Q.6.) Explain NFS client-server architecture.

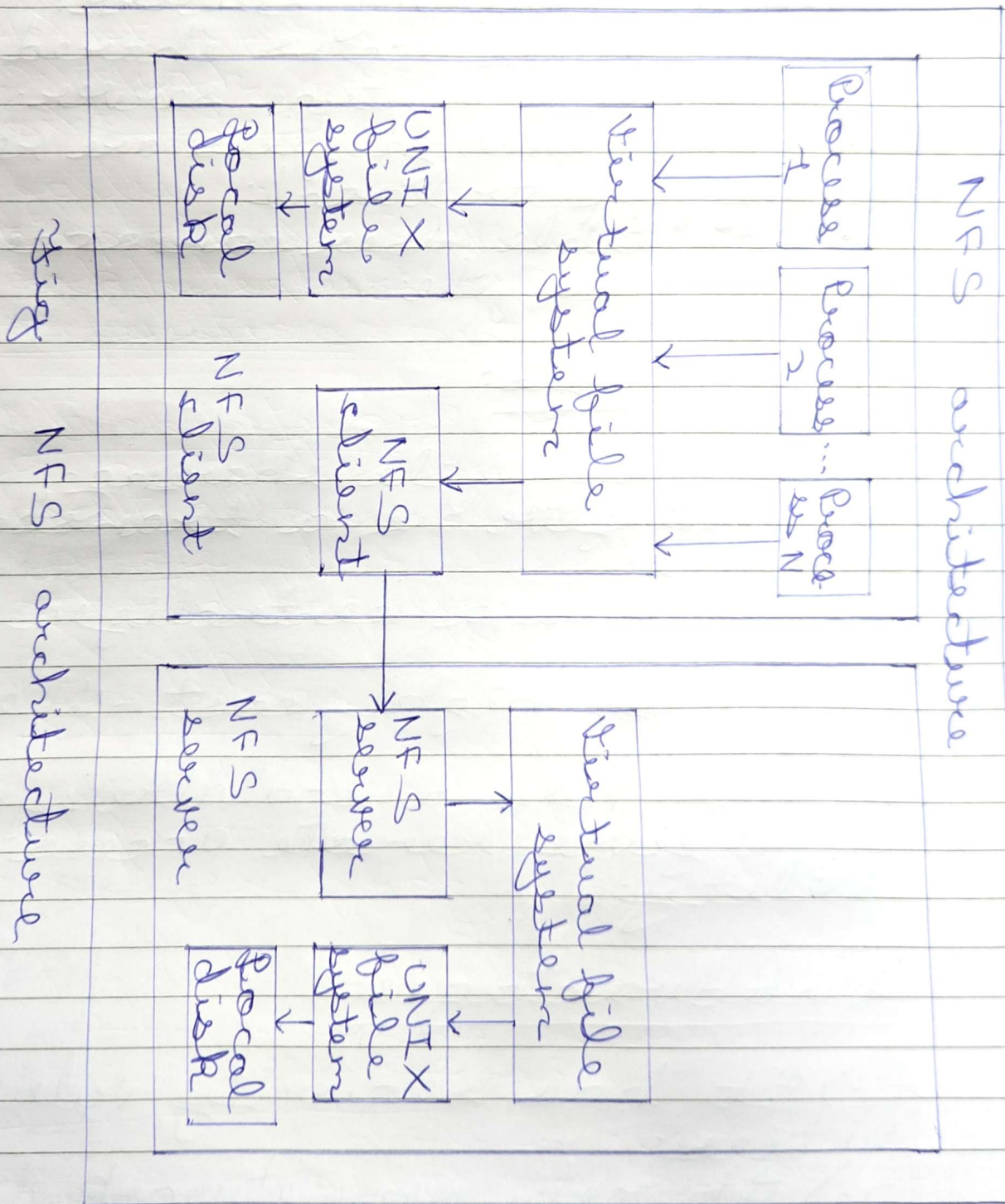
Ans. Definition:

NFS (Network File System) is a distributed file system that allows clients to access remote files as if they are stored locally.

Working:

- Client sends file access request.
- Server processes request.
- Now uses RPC (Remote Procedure Call).
- File data is returned to client.
- Client interacts with remote files transparently.

Diagram:



Example:

In a company network:

- Employees access shared files stored on a central server.
- Even though files are remote, they appear local.
- Multiple users can access the same file.

Advantages:

- 1.) Easy file sharing.
- 2.) It gives us a transparent access.
- 3.) It has a centralized storage.
- 4.) It is highly cost-effective.
- 5.) It has a scalable system.
- 6.) It gives remote access capability.

Disadvantages:

- 1.) It has security vulnerabilities.
- 2.) It relies on network depend-

dency.

3.) Risks of performance bottlenecks.

4.) There are server overload risks.

5.) Data inconsistency possible.

6.) It requires maintenance.

Q.7.) Explain security models with example.

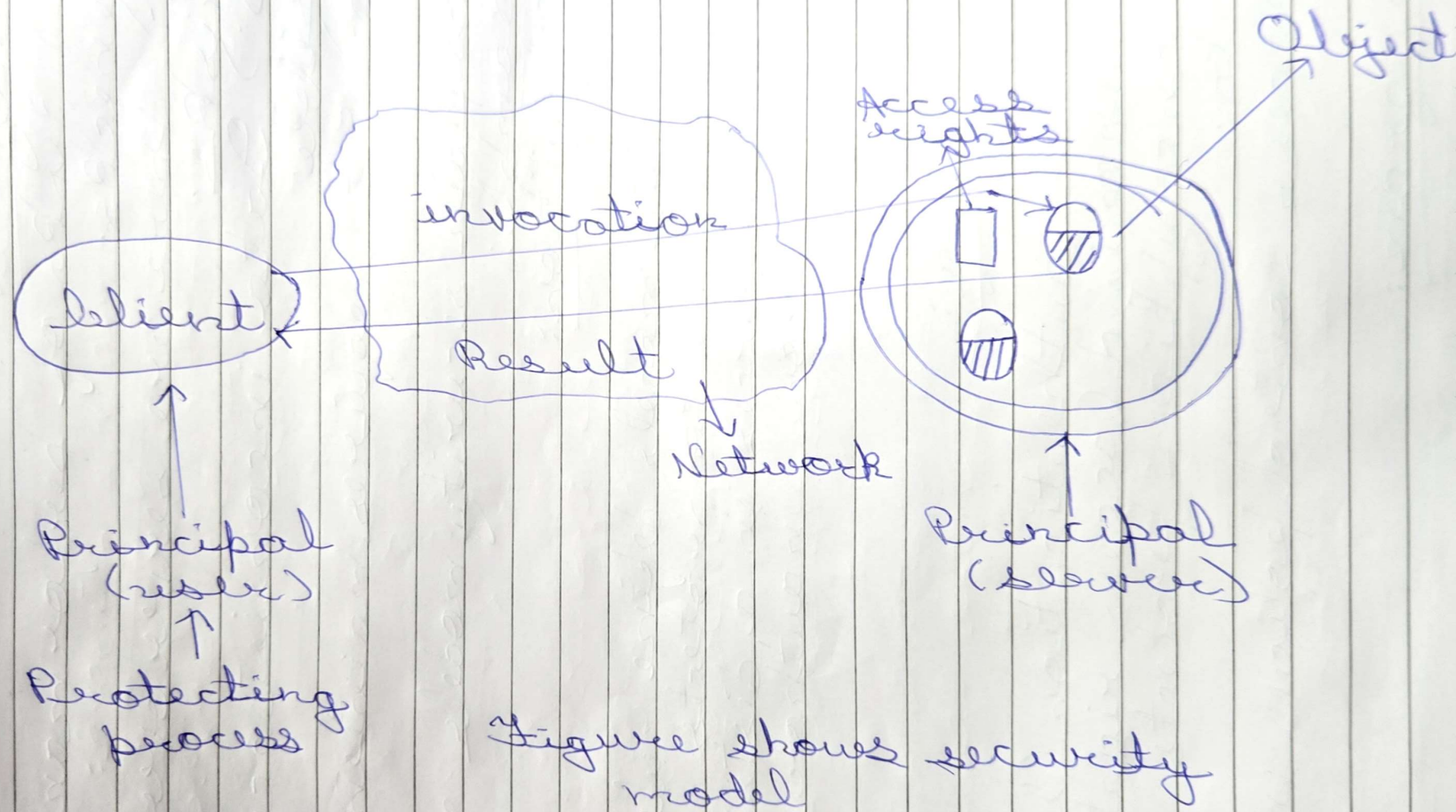
Ans Definition:

Security models define rules and mechanisms to protect data and resources in a distributed system from unauthorized access.

Working:

- User authentication (login).
- Authorized (access rights).
- Encryption protects the data.
- Access control policies enforced.

Diagram:



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Example:

1) Protects sensitive data:

In an online banking system:

- User logs in with password and OTP.
- Access right define what operations are allowed.
- Transactions are encrypted.
- Prevents unauthorized access.

Advantages:

- 1.) Prevents unauthorized access
- 2.) Ensures data integrity
- 3.) Protects sensitive data
- 4.) Build user trust
- 5.) Support legal compliance
- 6.) Enhances system security

Disadvantages:

- 1.) Implementation is very complex.
- 2.) Performance overhead is a big issue
- 3.) Infrastructure compared to others is costly.

4.) There can be user inconvenience.

5.) Maintenance is required.

6.) Risk of misconfiguration.

Q.8.) Explain message-oriented communication?

Ans Definition:

Message-oriented communication is a method where processes communicate by sending and receiving messages instead of sharing memory.

Working:

- Sender creates message.
- Message sent through network.
- Receiver receives it and processes it.
- Can be synchronous or asynchronous.
- Message is delivered as per request.

Diagram:

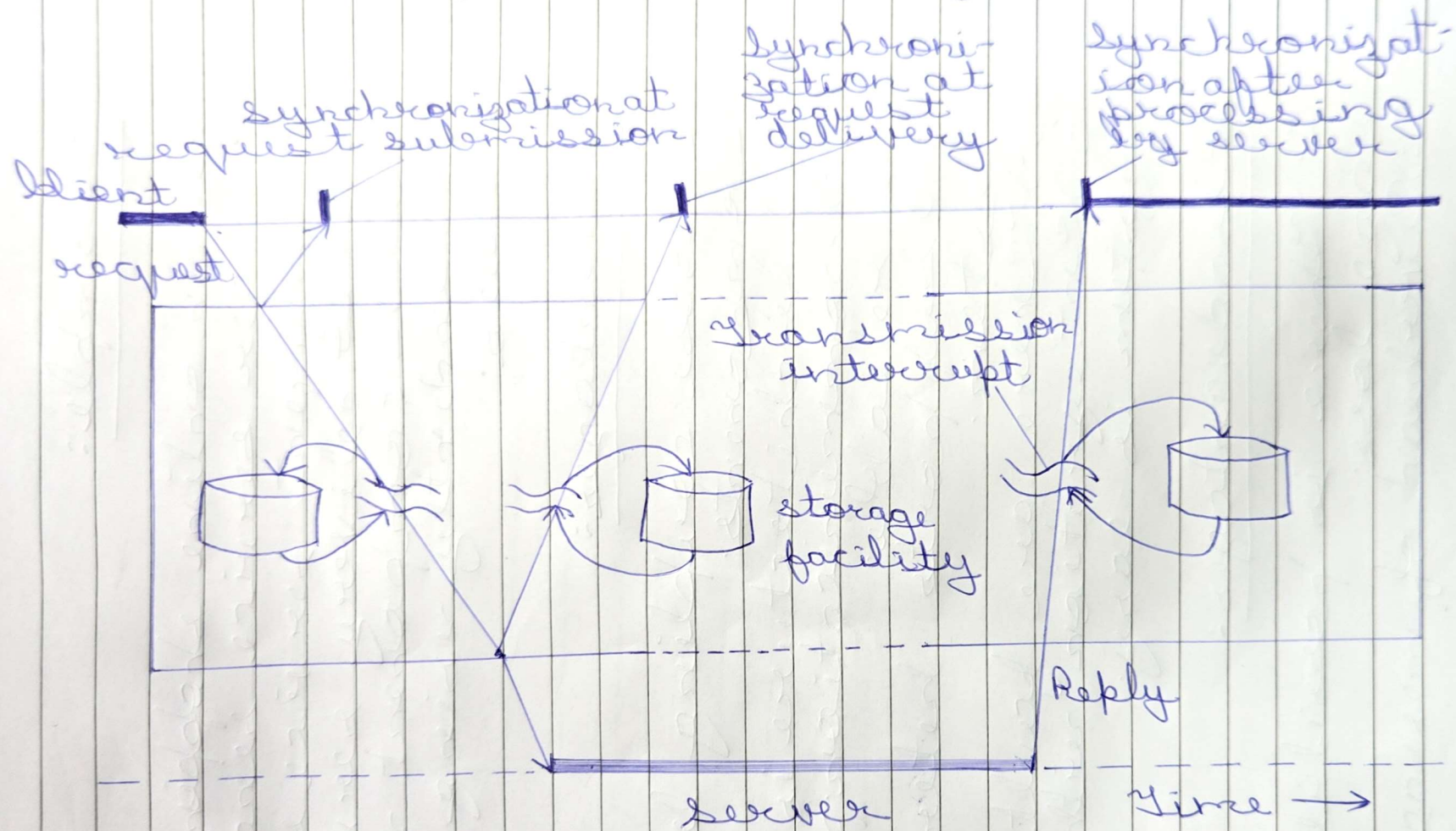


Fig. Message-oriented communication

Example:

In an online shopping system:

- Order service sends message.
- Payment service sends message to delivery service.
- Each component works independently using messages queues.

Advantages:

- 1.) Loose coupling between components.
- 2.) Architecture is scalable.
- 3.) Communication provided is reliable.
- 4.) Supports heterogeneous system.
- 5.) There are fault isolation.
- 6.) Integration with other system is easy.

Disadvantages:

- 1.) There is possibility of message loss.
- 2.) Overhead of message handling.